

Mobile Communication Networks

Lab Worksheets

2025/2026

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Worksheet n. 2

Wi-Fi and BLE Integration with Raspberry Pi Units

This laboratory work focuses on the experimental use of Wi-Fi and Bluetooth Low Energy (BLE) in a small wireless system based on Raspberry Pi units. In this worksheet, the emphasis is not on reproducing detailed configuration guides, but on implementing the system, performing measurements, observing its behavior, and discussing the results obtained.

The laboratory will be organized by islands, with each island shared by two groups of two students. Each island will have its own Wi-Fi access point/router infrastructure and two Raspberry Pi units for the experimental work, so that the groups can perform their measurements and BLE tests without depending on incompatible monitors or keyboards. The remaining equipment will be shared across the islands and the Wi-Fi configuration of each island will be done through the web interface of the assigned router or access point.

Note: Laboratory equipment will be shared among all groups throughout the weeks allocated for this assignment.

Part A — Initial Setup

A1 — Quick access to the Raspberry Pi units

Each group shall prepare two Raspberry Pi units, hereafter designated RPI1 and RPI2.

If the Raspberry Pi units are not already prepared, students should flash Raspberry Pi OS to the boot media before starting the lab. The recommended approach is to use Raspberry Pi Imager on a computer with an SD card reader, select Raspberry Pi OS, choose the storage device, enable SSH in the advanced/customization settings, define the username and password, and then write the image to a microSD card or, when supported by the model, to a USB drive or SSD.

When possible, SSH should be enabled during imaging so that the RPIs can be accessed remotely from the lab PC over Ethernet without requiring a local monitor or keyboard.

To avoid spending excessive time on operating system setup, students may follow these minimal steps:

- connect the RPI to the network through Ethernet and power it on;
- open an SSH terminal from the lab PC to each Raspberry Pi;
- identify the hostname and network interfaces using simple commands such as `hostname` and `ip a`;
- confirm that both units are operational.

Deliverable: table identifying RPI1 and RPI2, including hostname and active interfaces.

A2 — Wi-Fi connection of the two RPis

Using the Wi-Fi infrastructure assigned to the group, connect both RPis to the wireless network.

The group shall:

- verify that both devices are associated with the correct SSID;
- register the IP address obtained by each RPi;
- confirm IP connectivity between the two units and with the gateway.

Deliverable: table with SSID, IP addresses and image captures of connectivity confirmation.

Part B — Wi-Fi measurements between RPI1 and RPI2

B1 — Connectivity and latency

Using the Wi-Fi connection established in Part A, perform connectivity tests between RPI1 and RPI2.

The group shall:

- execute ``ping`` tests between the two RPis;
- repeat the measurement in at least two different physical conditions defined by the teacher, for example:
 - short distance with line of sight;
 - larger distance with one simple obstacle.
- register the average latency and any packet loss observed.

Deliverable: results table and short comment on the differences between the two conditions.

B2 — Throughput and link quality

Measure Wi-Fi performance between RPI1 and RPI2 using a simple throughput tool such as ``iperf3``, and register at least one signal quality indicator available in the Raspberry Pi.

The group shall:

- perform at least **three** throughput measurements in each test condition;
- register the corresponding signal information available, such as RSSI, link rate, or equivalent;
- whenever possible, repeat the tests in both 2.4 GHz and 5 GHz.

Deliverable: comparative table with throughput and link quality values, followed by a brief discussion of the observed behavior.

Part C — BLE connection between RPI1 and RPI2

C1 — BLE test with a smartphone

Using the Bluetooth/BLE interfaces of the Raspberry Pi units, start the BLE activity with a smartphone first, so that each group can verify the basic operation of the local BLE setup.

The group shall:

- activate the Bluetooth/BLE interface in the RPi assigned to the group;
- make the smartphone discoverable if needed and detect it from the Raspberry Pi;
- register device name/address information;
- establish a simple BLE connection with the smartphone and perform a short interaction, such as discovery, connection and one brief message/value exchange.

Deliverable: table with smartphone discovery results and captures of connection status.

C2 — Interoperability test between the two RPis

After the smartphone test has been validated, attempt BLE interoperability between the RPi of one group and the RPi of the other group in the same island, using a brief text message as the exchanged payload.

The group shall:

- establish the BLE link between the two RPis;
- exchange one short text message from one RPi to the other;
- repeat the test at least three times;
- register success/failure and any relevant observation;

Deliverable: short results table and brief comment.

Part D — Coexistence of Wi-Fi and BLE

With both RPis connected by Wi-Fi, repeat one of the Wi-Fi tests while a BLE operation is also active between the same devices.

The group shall:

- execute one Wi-Fi latency or throughput test;
- during the test, establish or use the BLE connection;
- compare the observations with the case where BLE is not active;
- indicate whether any visible impact on Wi-Fi or BLE behavior was observed.

Deliverable: short comparison table and technical comment.

The working groups

Groups of two students must be formed to answer the questions and prepare the report for this worksheet.

Report

A report including the deliverables must be uploaded to Moodle by the **12th of June of 2026**.

The report should include the percentage of effort contributed by each member of the group.

The presentation

Each group shall prepare a **10 minutes** presentation of their report deliverables.

After the presentation, 10 minutes will follow for questions and answers.

Each group member has to answer the questions addressed to him/her individually.

Groups will present their papers in the classroom on the **15, 16 and 17th of June of 2026**.